DATA ANALYTICS AND MACHINE LEARNING

Srijan Mukherjee

Selected Project Portfolio

Python | SQL | Power BI | Predictive modelling | Explainable AI

MSc Data Science and Artificial Intelligence, University of Liverpool
Expected completion January 2027 | Liverpool, UK

jon.mukerji@gmail.com | +44 7587 929262

GitHub portfolio

LinkedIn profile

Content opportunity scoring

FlyRank AI internship | Python, pandas, scikit-learn

Business question

Which pages should be prioritised for content refresh? I analysed 30,000 content-performance records to investigate declining pages and compare ways of ranking refresh opportunities.

Analysis and approach

Prepared the records for analysis and explored relationships between search volume, impressions, content length and performance. Investigated how click-through rate varied with search position. Trained logistic regression, decision-tree and random-forest models, then compared simple ranking rules with depth-limited trees.

Evaluation on held-out clients

Method	Precision at 20	Precision at 50
Hand rule	0.45	0.62
Depth 2 tree	0.60	0.56
Depth 3 tree	0.55	0.58

The depth 2 tree performed better among the top 20 selections; the hand rule performed better among the top 50. This comparison showed why model choice depends on the number of opportunities a team can act on. These results are from the client-holdout comparison, rather than training performance.

I also identified target leakage when a trend feature produced perfect Precision at 50. This reinforced the need to examine feature validity before trusting a high score.

View FlyRank analysis and notebooks

UK road safety analysis

Python | SQL | Power BI | Power Query | DAX

Analytical objective

Explore collision severity, time patterns and road conditions in UK road safety data, and organise the results into an interactive dashboard.

2025 dataset	Records
Collisions	101,525
Vehicles	183,948
Casualties	127,883

My contribution

Cleaned and analysed the collision, vehicle and casualty datasets. Built a four-page Power BI dashboard to explore severity, time patterns and road conditions and communicate the analysis through visual reporting.

Why this work matters

The project demonstrates handling related datasets at different levels of detail and presenting multiple views of a problem. For a business analyst, the same discipline supports clear reporting across users, transactions and products while keeping the unit of analysis explicit.

Browse project repositories on GitHub

SQL layoffs analysis

SQL | Data cleaning | Exploratory data analysis

Objective and delivery

Completed and published a SQL project covering data cleaning and exploratory analysis of layoffs data. The repository provides a work sample for reviewing my SQL approach and analysis.

Capability demonstrated

Using SQL to prepare data for analysis and investigate a dataset systematically. This complements the Python modelling and Power BI reporting work elsewhere in my portfolio.

View SQL project

Explainable Wi-Fi intrusion detection

MSc dissertation | AWID3 | Autoencoder | Random Forest | SHAP | LIME | Streamlit

Problem and implementation

Built a hybrid intrusion detection pipeline to identify enterprise Wi-Fi attacks, with a focus on Evil Twin detection. Combined an autoencoder, safety routing and Random Forest classification, and evaluated the pipeline on 5,564,048 held-out observations.

Reported held-out metric	Result
Attack recall	99.79%
False-negative rate	0.21%
Evil Twin F1 score	97.39%

Explainability and application

Implemented SHAP and LIME explanations to inspect model behaviour and built a Streamlit inference application. The work connected model development and evaluation with a usable interface and explanations.

Interpretation

These results describe the AWID3 held-out evaluation. They do not establish performance on live enterprise traffic. The transferable skills are anomaly investigation, classification, evaluation at scale and explaining predictions.

View dissertation implementation

Student placement prediction

Learning project | Python | Logistic regression

Built a binary classification model to predict student placement outcomes, practising train/test splitting and feature scaling. This project supports my understanding of a basic supervised learning workflow; it is a learning exercise rather than a validated recruitment decision tool.

Browse project repositories

Professional foundation

Around five years of technical support and network engineering experience at Enphase Energy, Infosys BPM, Cognizant, Concentrix and CSS Corp. This background strengthened my incident investigation, documentation and communication skills, including support for enterprise and banking infrastructure.